

CERTIFICATE OF ANALYSIS



SAMPLE INFORMATION

Sample Name: Month 3 - Refrigeration w/out moisture pack
Sample Matrix: Flower
Collected: 12/26/2018 10:02
Overall Result: **Pass**
Sample Id: 83710
Received: 12/26/2018 12:32

MANUFACTURER INFO

Business Name:
Street Address:
City:
Zip Code:

CANNABINOID ANALYSIS

- ❶ Total THC, CBD value(s) are corrected for moisture, and the Acids have been decarboxylated
- ❷ Total Cannabinoid value(s) are corrected for moisture content
- ❸ Individual Analyte value(s) are corrected for moisture content

TOTAL THC: 173.0 mg/g (17.30 %)
TOTAL CBD: ND (ND)
TOTAL CANNABINOIDS: 199.9 mg/g (19.99 %)

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
D9THC	11.68 mg/g (1.168 %)	0.2500	0.5000	D8THC	ND	0.2500	0.5000	
THCa	183.9 mg/g (18.39 %)	0.2500	0.5000	CBG	1.442 mg/g (0.1442 %)	0.2500	0.5000	
CBN	ND	0.2500	0.5000	CBD	ND	0.2500	0.5000	
CBDa	ND	0.2500	0.5000	CBGa	2.901 mg/g (0.2901 %)	0.2500	0.5000	
THCv	ND	0.2500	0.5000	CBC	ND	0.2500	0.5000	
CBDv	ND	0.2500	0.5000					

ADDITIONAL INFORMATION

Method: SOP-TECH-001
Instrument: HPLC-DAD

Sample Prepped 03/29/2019 12:03
Sample Analyzed 03/29/2019 18:30

Sample Approved 03/31/2019 14:51



CERTIFICATE OF ANALYSIS

TERPENE ANALYSIS

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Limonene	ND	0.0	0.0	Delta-3-Carene	ND	0.0	0.0	
Nerolidol	0.0010 mg/g	0.0	0.0	Linalool	ND	0.0	0.0	
Geraniol	ND	0.0	0.0	(-)-alpha-Bisabolol	0.0052 mg/g	0.0	0.0	
Gamma-Terpinene	ND	0.0	0.0	Ocimene	ND	0.0	0.0	
Terpinolene	ND	0.0	0.0	P-Cymene	ND	0.0	0.0	
Eucalyptol	ND	0.0	0.0	(-)-Isopulegol	ND	0.0	0.0	
Beta-Pinene	ND	0.0	0.0	Alpha-Pinene	0.0018 mg/g	0.0	0.0	
B-Myrcene	ND	0.0	0.0	Beta-Caryophyllene	0.0035 mg/g	0.0	0.0	
Alpha-Terpinene	ND	0.0	0.0	Alpha-Humulene	ND	0.0	0.0	
(-)-Guaiaol	ND	0.0	0.0	Camphene	ND	0.0	0.0	
(-)-Caryophyllene Oxide	ND	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-TECH-004
Instrument: HS-GC-FID

Sample Prepped 03/27/2019 17:11
Sample Analyzed 03/27/2019 19:34

Sample Approved 03/29/2019 08:37

MICROBIAL ANALYSIS

TEST TYPE RESULT: Pass
UNIT OF MEASUREMENT: Cycle Threshold (Ct)

ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
A.fumigatus	ND	0.0	0.0	33.00 Pass	A. flavus	ND	0.0	0.0	33.00 Pass
A. niger	ND	0.0	0.0	33.00 Pass	A. terreus	ND	0.0	0.0	33.00 Pass
STEC	ND	0.0	0.0	33.00 Pass	Salmonella spp	ND	0.0	0.0	33.00 Pass

ADDITIONAL INFORMATION

Method: SOP-LAB-008
Instrument: qPCR

Sample Prepped 03/29/2019 13:24
Sample Analyzed 04/01/2019 07:17

Sample Approved 04/01/2019 15:26





CERTIFICATE OF ANALYSIS

MOISTURE CONTENT

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Percent Moisture Content(%)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Moisture Content	11.94 %	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-LAB-006
Instrument: Moisture Analyzer

Sample Prepped 03/29/2019 14:09
Sample Analyzed 04/01/2019 19:41

Sample Approved 04/01/2019 19:49

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All LQC samples required by state regulations were performed and met the acceptance criteria.

DATA REVIEWED AND APPROVED BY

04/01/2019

Swetha Kaul, PhD
Chief Scientific Officer

Date



CERTIFICATE OF ANALYSIS



SAMPLE INFORMATION

Sample Name: Month 3 - Refrigeration w/ moisture pack
Sample Matrix: Flower
Collected: 12/26/2018 10:00
Overall Result: **Pass**
Sample Id: 83709
Received: 12/26/2018 12:32

MANUFACTURER INFO

Business Name:
Street Address:
City:
Zip Code:

CANNABINOID ANALYSIS

- ❶ Total THC, CBD value(s) are corrected for moisture, and the Acids have been decarboxylated
- ❷ Total Cannabinoid value(s) are corrected for moisture content
- ❸ Individual Analyte value(s) are corrected for moisture content

TOTAL THC: 183.1 mg/g (18.31 %)
TOTAL CBD: ND (ND)
TOTAL CANNABINOIDS: 211.4 mg/g (21.14 %)

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
D9THC	13.04 mg/g (1.304 %)	0.2500	0.5000	D8THC	ND	0.2500	0.5000	
THCa	193.9 mg/g (19.39 %)	0.2500	0.5000	CBG	1.430 mg/g (0.1430 %)	0.2500	0.5000	
CBN	ND	0.2500	0.5000	CBD	ND	0.2500	0.5000	
CBDa	ND	0.2500	0.5000	CBGa	3.053 mg/g (0.3053 %)	0.2500	0.5000	
THCv	ND	0.2500	0.5000	CBC	ND	0.2500	0.5000	
CBDv	ND	0.2500	0.5000					

ADDITIONAL INFORMATION

Method: SOP-TECH-001
Instrument: HPLC-DAD

Sample Prepped 03/29/2019 12:02
Sample Analyzed 03/29/2019 18:30

Sample Approved 03/31/2019 14:51



CERTIFICATE OF ANALYSIS

TERPENE ANALYSIS

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Limonene	0.0273 mg/g	0.0	0.0	Delta-3-Carene	ND	0.0	0.0	
Nerolidol	0.0046 mg/g	0.0	0.0	Linalool	0.0203 mg/g	0.0	0.0	
Geraniol	0.0208 mg/g	0.0	0.0	(-)-alpha-Bisabolol	0.0220 mg/g	0.0	0.0	
Gamma-Terpinene	ND	0.0	0.0	Ocimene	ND	0.0	0.0	
Terpinolene	ND	0.0	0.0	P-Cymene	0.0036 mg/g	0.0	0.0	
Eucalyptol	ND	0.0	0.0	(-)-Isopulegol	ND	0.0	0.0	
Beta-Pinene	0.0397 mg/g	0.0	0.0	Alpha-Pinene	0.0857 mg/g	0.0	0.0	
B-Myrcene	0.0417 mg/g	0.0	0.0	Beta-Caryophyllene	0.0754 mg/g	0.0	0.0	
Alpha-Terpinene	ND	0.0	0.0	Alpha-Humulene	0.0314 mg/g	0.0	0.0	
(-)-Guaiaol	ND	0.0	0.0	Camphene	0.0031 mg/g	0.0	0.0	
(-)-Caryophyllene Oxide	ND	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-TECH-004
Instrument: HS-GC-FID

Sample Prepped 03/27/2019 17:11
Sample Analyzed 03/27/2019 19:34

Sample Approved 03/29/2019 08:37

MICROBIAL ANALYSIS

TEST TYPE RESULT: Pass
UNIT OF MEASUREMENT: Cycle Threshold (Ct)

ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
A.fumigatus	ND	0.0	0.0	33.00 Pass	A. flavus	ND	0.0	0.0	33.00 Pass
A. niger	ND	0.0	0.0	33.00 Pass	A. terreus	ND	0.0	0.0	33.00 Pass
STEC	ND	0.0	0.0	33.00 Pass	Salmonella spp	ND	0.0	0.0	33.00 Pass

ADDITIONAL INFORMATION

Method: SOP-LAB-008
Instrument: qPCR

Sample Prepped 03/29/2019 13:24
Sample Analyzed 04/01/2019 07:17

Sample Approved 04/01/2019 15:26





CERTIFICATE OF ANALYSIS

MOISTURE CONTENT

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Percent Moisture Content(%)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Moisture Content	13.65 %	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-LAB-006
Instrument: Moisture Analyzer

Sample Prepped 03/29/2019 14:07
Sample Analyzed 04/01/2019 19:41

Sample Approved 04/01/2019 19:49

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DATA REVIEWED AND APPROVED BY

04/01/2019

Swetha Kaul, PhD
Chief Scientific Officer

Date



CERTIFICATE OF ANALYSIS



SAMPLE INFORMATION

Sample Name: Month 3 - Ambient w/out moisture pack
Sample Matrix: Flower
Collected: 12/26/2018 09:59
Overall Result: **Pass**
Sample Id: 83708
Received: 12/26/2018 12:32

MANUFACTURER INFO

Business Name:
Street Address:
City:
Zip Code:

CANNABINOID ANALYSIS

- ❶ Total THC, CBD value(s) are corrected for moisture, and the Acids have been decarboxylated
- ❷ Total Cannabinoid value(s) are corrected for moisture content
- ❸ Individual Analyte value(s) are corrected for moisture content

TOTAL THC: 174.0 mg/g (17.40 %)
TOTAL CBD: ND (ND)
TOTAL CANNABINOIDS: 201.2 mg/g (20.12 %)

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
D9THC	14.47 mg/g (1.447 %)	0.2500	0.5000	D8THC	ND	0.2500	0.5000	
THCa	181.9 mg/g (18.19 %)	0.2500	0.5000	CBG	1.571 mg/g (0.1571 %)	0.2500	0.5000	
CBN	ND	0.2500	0.5000	CBD	ND	0.2500	0.5000	
CBDa	ND	0.2500	0.5000	CBGa	3.230 mg/g (0.3230 %)	0.2500	0.5000	
THCv	ND	0.2500	0.5000	CBC	ND	0.2500	0.5000	
CBDv	ND	0.2500	0.5000					

ADDITIONAL INFORMATION

Method: SOP-TECH-001
Instrument: HPLC-DAD

Sample Prepped 03/29/2019 12:00
Sample Analyzed 03/29/2019 18:30

Sample Approved 03/31/2019 14:51



CERTIFICATE OF ANALYSIS

TERPENE ANALYSIS

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Limonene	0.0318 mg/g	0.0	0.0	Delta-3-Carene	ND	0.0	0.0	
Nerolidol	0.0088 mg/g	0.0	0.0	Linalool	0.0221 mg/g	0.0	0.0	
Geraniol	0.0207 mg/g	0.0	0.0	(-)-alpha-Bisabolol	0.0299 mg/g	0.0	0.0	
Gamma-Terpinene	0.0022 mg/g	0.0	0.0	Ocimene	ND	0.0	0.0	
Terpinolene	ND	0.0	0.0	P-Cymene	0.0035 mg/g	0.0	0.0	
Eucalyptol	ND	0.0	0.0	(-)-Isopulegol	ND	0.0	0.0	
Beta-Pinene	0.0456 mg/g	0.0	0.0	Alpha-Pinene	0.1048 mg/g	0.0	0.0	
B-Myrcene	0.0490 mg/g	0.0	0.0	Beta-Caryophyllene	0.0769 mg/g	0.0	0.0	
Alpha-Terpinene	ND	0.0	0.0	Alpha-Humulene	0.0295 mg/g	0.0	0.0	
(-)-Guaiaol	ND	0.0	0.0	Camphene	0.0035 mg/g	0.0	0.0	
(-)-Caryophyllene Oxide	ND	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-TECH-004
Instrument: HS-GC-FID

Sample Prepped 03/27/2019 17:11
Sample Analyzed 03/27/2019 19:34

Sample Approved 03/29/2019 08:37

MICROBIAL ANALYSIS

TEST TYPE RESULT: Pass
UNIT OF MEASUREMENT: Cycle Threshold (Ct)

ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
A.fumigatus	ND	0.0	0.0	33.00 Pass	A. flavus	ND	0.0	0.0	33.00 Pass
A. niger	ND	0.0	0.0	33.00 Pass	A. terreus	ND	0.0	0.0	33.00 Pass
STEC	ND	0.0	0.0	33.00 Pass	Salmonella spp	ND	0.0	0.0	33.00 Pass

ADDITIONAL INFORMATION

Method: SOP-LAB-008
Instrument: qPCR

Sample Prepped 03/29/2019 13:23
Sample Analyzed 04/01/2019 07:17

Sample Approved 04/01/2019 15:26





CERTIFICATE OF ANALYSIS

MOISTURE CONTENT

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Percent Moisture Content(%)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Moisture Content	12.06 %	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-LAB-006
Instrument: Moisture Analyzer

Sample Prepped 03/29/2019 14:05
Sample Analyzed 04/01/2019 19:41

Sample Approved 04/01/2019 19:49

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DATA REVIEWED AND APPROVED BY

04/01/2019

Swetha Kaul, PhD
Chief Scientific Officer

Date



CERTIFICATE OF ANALYSIS



SAMPLE INFORMATION

Sample Name: Month 3 - Ambient w/ moisture pack
Sample Matrix: Flower
Collected: 12/26/2018 09:59
Overall Result: **Pass**
Sample Id: 83707
Received: 12/26/2018 12:32

MANUFACTURER INFO

Business Name:
Street Address:
City:
Zip Code:

CANNABINOID ANALYSIS

- ❶ Total THC, CBD value(s) are corrected for moisture, and the Acids have been decarboxylated
- ❷ Total Cannabinoid value(s) are corrected for moisture content
- ❸ Individual Analyte value(s) are corrected for moisture content

TOTAL THC: 176.2 mg/g (17.62 %)
TOTAL CBD: ND (ND)
TOTAL CANNABINOIDS: 201.9 mg/g (20.19 %)

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
D9THC	14.77 mg/g (1.477 %)	0.2500	0.5000	D8THC	ND	0.2500	0.5000	
THCa	184.1 mg/g (18.41 %)	0.2500	0.5000	CBG	ND	0.2500	0.5000	
CBN	ND	0.2500	0.5000	CBD	ND	0.2500	0.5000	
CBDa	ND	0.2500	0.5000	CBGa	3.043 mg/g (0.3043 %)	0.2500	0.5000	
THCv	ND	0.2500	0.5000	CBC	ND	0.2500	0.5000	
CBDv	ND	0.2500	0.5000					

ADDITIONAL INFORMATION

Method: SOP-TECH-001
Instrument: HPLC-DAD

Sample Prepped 03/29/2019 11:27
Sample Analyzed 03/29/2019 18:30

Sample Approved 03/31/2019 14:51



CERTIFICATE OF ANALYSIS

TERPENE ANALYSIS

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Milligrams per Gram(mg/g)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Limonene	0.0253 mg/g	0.0	0.0	Delta-3-Carene	ND	0.0	0.0	
Nerolidol	0.0060 mg/g	0.0	0.0	Linalool	0.0197 mg/g	0.0	0.0	
Geraniol	0.0193 mg/g	0.0	0.0	(-)-alpha-Bisabolol	0.0146 mg/g	0.0	0.0	
Gamma-Terpinene	ND	0.0	0.0	Ocimene	ND	0.0	0.0	
Terpinolene	ND	0.0	0.0	P-Cymene	0.0033 mg/g	0.0	0.0	
Eucalyptol	0.0009 mg/g	0.0	0.0	(-)-Isopulegol	ND	0.0	0.0	
Beta-Pinene	0.0394 mg/g	0.0	0.0	Alpha-Pinene	0.0915 mg/g	0.0	0.0	
B-Myrcene	0.0382 mg/g	0.0	0.0	Beta-Caryophyllene	0.0620 mg/g	0.0	0.0	
Alpha-Terpinene	ND	0.0	0.0	Alpha-Humulene	0.0280 mg/g	0.0	0.0	
(-)-Guaio	ND	0.0	0.0	Camphene	0.0031 mg/g	0.0	0.0	
(-)-Caryophyllene Oxide	ND	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-TECH-004
Instrument: HS-GC-FID

Sample Prepped 03/27/2019 17:11
Sample Analyzed 03/27/2019 19:34

Sample Approved 03/29/2019 08:37

MICROBIAL ANALYSIS

TEST TYPE RESULT: Pass
UNIT OF MEASUREMENT: Cycle Threshold (Ct)

ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
A.fumigatus	ND	0.0	0.0	33.00 Pass	A. flavus	ND	0.0	0.0	33.00 Pass
A. niger	ND	0.0	0.0	33.00 Pass	A. terreus	ND	0.0	0.0	33.00 Pass
STEC	ND	0.0	0.0	33.00 Pass	Salmonella spp	ND	0.0	0.0	33.00 Pass

ADDITIONAL INFORMATION

Method: SOP-LAB-008
Instrument: qPCR

Sample Prepped 03/29/2019 13:22
Sample Analyzed 04/01/2019 07:17

Sample Approved 04/01/2019 15:26





CERTIFICATE OF ANALYSIS

MOISTURE CONTENT

TEST TYPE RESULT: N/A
UNIT OF MEASUREMENT: Percent Moisture Content(%)

ANALYTE	RESULT	LOD	LLOQ	ANALYTE	RESULT	LOD	LLOQ	ACTION LEVEL
Moisture Content	13.97 %	0.0	0.0					

ADDITIONAL INFORMATION

Method: SOP-LAB-006
Instrument: Moisture Analyzer

Sample Prepped 03/29/2019 14:03
Sample Analyzed 04/01/2019 19:41

Sample Approved 04/01/2019 19:49

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04/01/2019

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